Ling 285

Exam 1 Review, Questions will be answered in Tuesday 9/29 Class

1. What's the similarity between "m" in "mad" and "b" in "bad", and what's the difference between those sounds?

2. What's the role of the learning rate in Stochastic Gradient Descent and the role of stiffness in (viscous) springy systems? What happens when learning rate is high vs. low and when stiffness is high vs. low?

3. What's the physical principle behind how your auditory system hears different frequencies? Describe the basic physics of the basilar membrane to illustrate how the system works.

4. Imagine the following two temperature control systems, if you set the temperature at 70: a) Excellent thermostat right near the Air conditioning/Heating Unit: If the temperature is higher than 70, the AC turns on and temperature goes down to 70 and stays there, as the thermostat detects the right temperature and stops the AC. If the temperature were lower than 70, heating turns on till it gets to 70, and thermostat turns it off as it detects the right temperature. b) Thermostat really far from the AC/HU: If the temperature is higher than 70, the AC turns on and temperature goes down to 70, but the thermostat doesn't detect that and the temperature keeps dropping till 60, and when the thermostat finally detects 60, it turns on HU, and heat comes on, and temperature rises to 70, but thermostat doesn't detect it, till the temperature goes up to 80, etc. Which of these systems is like a spring by itself, and which is like a mass-spring?

5. Why is the exhaustive algorithm so inefficient for solving the shortest path problem?

6. What are the different ways we distinguish between the consonants at the beginning of the words: Pam, Bam, Dam, Tam, Kam, and Gam?

7. Describe the psychological principle of closure, and how it's used in the GPT system?

8. How is knowledge represented in Neural Networks?

9. What's the basic tradeoff that we see in both shortest path problems and Stochastic Gradient Descent Algorithm?

10. What's my favorite popular way of describe how learning in AI works? Fake __ ___ ___ ___ ____. Describe why I describe it this way.

11. What's the output of an LLM or Neural Network in general, and how does it express a decision?

12. Is this a good output of an NN, if there are 4 categories [.4,0,.5,.2]? Why yes/no?

13. In each vowel, air vibrates at a single frequency? Yes/No

14. What are the 2 basic ways we distinguish vowels?

15. Some people say “Everything is a Harmonic Oscillator”. Give 5 examples of oscillatory systems, we’ve discussed in class and/or in the Meadows book.

16. If $\dot{x} = v$
 $\dot{v} = -\frac{k}{m} x$ starts with x negative and v = 0, will it oscillate? Say in a couple of sentences, what will happen.

17. If $\dot{x} = -.1\,x$ describes a system, then if x starts negative, what will happen? What happens if the number $-.1$ becomes $-.9$?

18. In SGD for averaging a list, the mathematical formula for the New Estimate of average x is: New x = Current x – lr (Current x – data point d to learn from). What is the purpose of the first minus in the formula? To answer, consider the possibility that the data point is above vs. below the Current x.

19. What is the math of forward propagation from the signal input to an NN all the way to a decision at the end? Describe it.

20. Imagine that the decision by an NN is [.1,.1,.1,.2,.5], but the true category is the one in the 2th position, using Python list notation, then how do we express to the NN in a probability distribution what the true answer is?